STAT 618
Bayesian Statistics

Quiz # 2

February 7, 2018

Name: SOLUTION

There is a 40% chance of rain, a 10% chance of snow, and a 50% chance of no precipitation. Mr. X decides between driving to work or taking subway. His loss function is the amount of time his chosen route takes. He knows that:

– Under dry conditions, driving to work takes 20 minutes;
– During rain, driving takes 30 minutes;
– During snow, driving takes 50 minutes;
– Subway always takes 37 minutes from home to work.

(a) Help Mr. X find the Bayes decision on whether to drive or to take subway.
(b) Help Mr. X find the minimax decision.
(c) Which decisions are admissible?

SOLUTION. We are given the following loss matrix and prior distribution.

Parameter θ	Prior $\pi(\theta)$	Decisions: δ_1, drive	Decisions: δ_2, subway
θ_1, rain	0.4	30	37
θ_2, snow	0.1	50	37
θ_3, dry	0.5	20	37

(a) Compute the Bayes risk for each decision. This is the prior risk, because there is no data in this problem.

$$\begin{aligned}\rho(\pi, \delta_1) &= (0.4)(30) + (0.1)(50) + (0.5)(20) = 27 \\ \rho(\pi, \delta_1) &= 37\end{aligned}$$

The Bayes decision is $\underline{\delta_1\text{, to drive}}$.

(b) Compute the maximum risk for each decision.

$$\begin{aligned}\sup_\theta \rho(\pi, \delta_1) &= 50 \\ \sup_\theta \rho(\pi, \delta_1) &= 37\end{aligned}$$

The minimax decision is $\underline{\delta_2\text{, to take subway}}$.

(c) δ_1 is not R-better than δ_2 because $L(\theta_2, \delta_1) > L(\theta_2, \delta_2)$.
δ_2 is not R-better than δ_1 because $L(\theta_1, \delta_2) > L(\theta_1, \delta_1)$.
Therefore, both decisions are admissible.